

CAR DAMAGE DETECTION

Internship Project Report

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Car Damage Detection Task

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1. Project Scope

As part of the internship, a computer vision project was carried out for detecting exterior car damage from images.

The main task was to investigate image-based car damage detection and develop a model that could detect two target damage types:

- dent
- scratch

The project focused particularly on small and subtle dents and scratches rather than severe or large-scale damage.

During the initial stage, research was conducted on computer vision, Object Detection, and YOLO-based approaches. Different datasets and model options were also investigated to determine an appropriate approach for the task.

2. Research and Dataset Preparation

The first stage of the project focused on finding and preparing suitable car damage datasets. Multiple datasets were examined because their class structures, class names, and damage categories were different.

The datasets were combined and then reviewed to create a consistent structure for the project. Classes outside the project scope were removed, while relevant classes were consolidated into the two target classes:

- dent
- scratch

The dataset preparation process was carried out using **Roboflow**.

The final dataset contains:

Split	Images
Train	16,331
Validation	2,041
Test	2,042
Total	20,414

There are **40,616 annotated instances** in total:

Class	Annotations
dent	15,395
scratch	25,221
Total	40,616

The final dataset was organized in YOLO format with two classes:

0 → dent

1 → scratch

The dataset was divided into training, validation, and test sets and prepared for the model experiments.

3. Model Experiments

After the dataset preparation stage, supervised Object Detection approaches were investigated and YOLO models were used for the experiments.

First, **YOLOv8n** was trained as a baseline. This experiment used an earlier version of the merged dataset, before the final cleaning and restructuring process.

Following the baseline experiment, the final cleaned dataset was used to train:

- **YOLO11n**
- **YOLO11s**

The model training experiments were performed in **Google Colab** using a GPU environment.

YOLOv8n was trained for 50 epochs with a 640×640 image size and batch size 16.

YOLO11n and YOLO11s were trained for 50 epochs with a 640×640 image size and automatic batch size. Pretrained weights and the automatic optimizer configuration were used.

The models were evaluated using:

- Precision
- Recall
- mAP50
- mAP50-95

4. Test and Evaluation

After training, the models were evaluated separately on the unseen test set containing **2,042 images and 3,741 annotated instances**.

The test results were:

Model	Precision	Recall	mAP50	mAP50-95
YOLOv8n	20.88%	14.89%	11.03%	5.79%
YOLO11n	72.39%	53.59%	60.44%	36.31%
YOLO11s	72.45%	51.77%	59.03%	35.50%

The YOLO11n and YOLO11s experiments used the final cleaned dataset, while YOLOv8n was originally trained using an earlier dataset version. Therefore, the results were evaluated with this dataset difference in mind.

Class-level results were also examined. For the YOLO11 experiments, **dent** detection produced higher Recall and mAP values than **scratch** detection. This indicated that detecting scratches remained a more challenging part of the task.

In addition to numerical metrics, confusion matrices, precision-recall curves, and sample prediction images were generated to examine the model outputs.

5. Demo Application

After the model experiments, an interactive demo application was developed to demonstrate the inference process.

The application was developed using **Streamlit**. Users can upload a car image and run the trained model on the image.

The application displays:

- Detected damage type
- Bounding boxes
- Confidence scores

The demo uses the trained **YOLO11n** model. The trained model weights are hosted separately on **Hugging Face**, while the application code and project files are maintained on **GitHub**.

The application was deployed using **Streamlit Community Cloud**, allowing the model to be tested through a web interface.

6. Conclusion

Within the scope of the internship task, a complete workflow for image-based car damage detection was investigated and implemented.

The work included researching Object Detection and YOLO-based approaches, collecting and reviewing datasets, cleaning and restructuring the annotations, preparing the final dataset, training different YOLO models, evaluating the trained models on an unseen test set, and examining the resulting predictions.

Finally, a Streamlit-based demo was prepared to demonstrate the trained model on user-uploaded car images. The project code, documentation, experiment results, and evaluation outputs were organized in a GitHub repository.

Project Links and Resources

GitHub Repository

Car Damage Detection project repository containing the project code, documentation, experiment results, and evaluation outputs.

Live Demo

Streamlit-based web application for testing the trained YOLO11n model.

Hugging Face Model

The trained YOLO11n model weights (`best.pt`) are hosted separately on Hugging Face.

Roboflow – Original Merged Dataset

Dataset used during the initial dataset preparation and YOLOv8n baseline experiment.

Roboflow – Final Cleaned Dataset

Final dataset containing the two target classes, `dent` and `scratch`.